Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

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gem5 TimingSimpleCPU · workload profile `train` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, direct-mapped L1D/L2 · gem5 `e0a12aa4e3+dirty`

Definitions (metric schema 9). Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` ≤ `store` and `load` ≥ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

Benchmark	ROI_insts	stack_event	**spill_event**	csave_event	other_event	spill/stack %	Spill %	Csave %	L1D m %	L2 m %	sp_load m %	sp_store m %	cs_load m %	cs_store m %
cactuBSSN	383 945 680 437	30 560 138 315	30 377 509 662	181 932 059	696 594	99.4	21.33	0.09	17.365	48.81	6.420	5.496	1.484	0.271
parest	299 200 615 391	580 499 665	94 157 487	462 351 877	23 990 301	16.2	0.18	0.31	15.291	50.52	13.828	5.544	4.604	5.835
deepsjeng	399 987 651 203	14 214 822 378	3 538 346 146	10 408 044 378	268 431 854	24.9	2.99	5.20	13.572	20.58	6.641	9.693	3.991	4.802
leela	436 316 202 851	13 338 709 984	512 185 059	12 317 582 149	508 942 776	3.8	0.43	5.65	7.771	15.93	12.088	0.225	3.153	1.900
namd	218 427 292 248	2 350 200 000	2 339 066 714	8 976 130	2 157 156	99.5	3.22	0.01	15.525	37.27	11.949	2.502	4.394	2.397
lbm	100 160 472 568	4 229 647 076	4 229 634 900	11 332	844	100.0	9.69	0.00	26.827	74.99	1.629	5.336	10.678	1.209
nab	38 232 521 997	784 372 830	637 890 733	128 504 979	17 977 118	81.3	7.61	0.67	11.611	16.55	4.174	3.096	3.381	3.202
nab_gcn4dna	366 315 945 523	7 770 025 698	5 788 787 408	1 746 098 454	235 139 836	74.5	7.30	0.95	11.338	31.28	4.327	3.986	3.149	3.343
xz	45 071 901 345	1 313 479 362	149 252 420	1 077 227 617	86 999 325	11.4	1.36	4.78	8.854	32.06	3.589	9.751	1.675	2.574
xz_combined	103 539 847 907	1 438 295 106	686 344 414	673 142 677	78 808 015	47.7	2.79	1.30	14.676	25.02	7.094	5.232	5.413	6.019
imagick	284 114 941 975	294 947 576	3 377 502	291 114 097	455 977	1.1	0.04	0.20	5.624	86.53	28.947	2.476	0.090	0.304
blender	721 231 649 060	11 980 515 650	494 531 875	10 340 415 008	1 145 568 767	4.1	0.31	2.87	9.957	27.63	6.675	7.114	5.582	6.198
povray	28 621 640 749	2 169 279 970	27 550 868	2 002 206 555	139 522 547	1.3	0.27	13.99	14.296	36.10	12.024	21.547	4.564	4.442
omnetpp	134 781 950 948	8 710 005 719	5 235 206	8 476 934 111	227 836 402	0.1	0.01	12.58	18.539	41.67	0.723	10.689	4.354	4.020
xalancbmk	270 346 763 301	25 872 371 471	370 722 439	24 133 124 322	1 368 524 710	1.4	0.44	17.85	14.578	26.71	30.232	25.465	3.104	4.949
x264	386 397 511 704	16 699 463 061	11 425 519 865	4 723 224 155	550 719 041	68.4	6.30	2.44	4.045	26.31	0.928	1.200	1.720	2.173
gcc_200	118 445 469 526	9 732 539 869	435 131 547	9 187 999 599	109 408 723	4.5	1.16	15.51	12.949	37.34	7.301	8.798	2.693	3.571
gcc_scilab	92 316 903 709	7 134 405 901	365 548 253	6 690 997 652	77 859 996	5.1	1.25	14.50	13.633	38.00	7.843	8.196	2.770	3.710
gcc	10 027 112 139	797 875 479	42 973 169	745 789 454	9 112 856	5.4	1.41	14.88	13.466	39.11	6.700	8.236	2.821	3.814
mcf	125 387 999 079	1 200 460 284	483 048 594	714 151 251	3 260 439	40.2	1.61	1.14	26.678	71.66	10.486	10.536	5.620	6.519
perlbench_scrabbl	29 794 284 033	2 637 314 691	19 766 463	2 595 400 347	22 147 881	0.7	0.23	17.42	12.387	16.92	5.421	15.503	7.057	4.858
perlbench_suns	1 489 955 308	141 093 681	4 683 473	133 092 545	3 317 663	3.3	0.98	17.87	9.862	25.17	2.953	10.611	3.530	2.782
perlbench_diffmail	71 841 738 045	5 980 785 368	357 324 444	5 559 068 861	64 392 063	6.0	2.06	15.48	10.331	26.66	2.391	10.536	3.199	3.979
perlbench_perfect	12 407 457 725	1 157 150 004	57 361 433	1 079 863 927	19 924 644	5.0	1.44	17.41	11.060	49.94	1.540	7.692	1.660	2.386
perlbench_splitmail	36 465 452 492	2 676 936 460	458 223 604	2 146 373 965	72 338 891	17.1	5.79	11.77	8.886	21.52	3.081	2.969	3.127	2.272
bwaves_1	94 269 943 053	669 560 992	669 492 761	55 690	12 541	100.0	2.99	0.00	43.947	26.39	6.178	1.303	4.893	5.680
bwaves_2	91 219 148 006	659 168 858	659 100 916	55 475	12 467	100.0	3.04	0.00	43.905	26.65	6.082	1.324	4.827	5.666
exchange2	177 369 845 254	5 445 875 346	3 838 435 663	1 053 305 830	554 133 853	70.5	4.84	1.19	3.655	1.48	2.783	2.443	2.474	3.955
fotonik3d	188 709 212 304	1 778 874 318	1 763 902 508	13 601 309	1 370 501	99.2	3.99	0.01	22.468	36.82	9.882	0.425	0.760	3.970
wrf	519 187 794 216	2 589 068 129	681 696 831	1 750 905 769	156 465 529	26.3	0.92	0.67	25.818	26.35	13.717	8.463	5.685	2.616
cam4	63 971 148 372	1 115 718 649	416 171 917	673 858 878	25 687 854	37.3	2.49	2.11	14.713	46.92	5.035	3.661	2.137	2.781
Total	5 849 596 052 468	186 023 601 890	**70 932 974 274**	109 315 410 452	5 775 217 164	38.1	3.60	3.74	0.000	—	—	—	—	—